

ORIGINAL RESEARCH

Interval prediction of PV power based on double sample entropy framework and integrated bidomain decomposition deep learning

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Abstract

A method for photovoltaic (PV) power interval prediction based on double sample entropy framework and integrated bidomain decomposition deep learning is proposed to address the volatility, randomness, intermittency and timeliness of PV power prediction. By using the integrated bidomain decomposition method under the framework of double sample entropy, the PV power time series is decomposed into several components with similar complexity and the deep learning network is used to learn and predict the results of each component reconstruction to obtain point predictions. The quantile regression method is then used based on the point prediction to obtain interval predictions. The actual calculation examples show that the RMSE and CPRS of the proposed method are reduced by an average of 45.56% and 31.42%, respectively, compared with other integrated bidomain decomposition deep learning methods that are not in the double sample entropy framework and single decomposition deep learning methods.

1 | INTRODUCTION

In recent years, the world's energy consumption structure has undergone transformation, with the proportion of renewable energy in power generation continuing to rise. According to the International Energy Agency (IEA) market analysis and forecast from 2019 to 2024 on renewable energy and technologies, renewable power capacity is set to expand by 50% [1]. Photovoltaic (PV) power generation is one of the ideal applications of renewable energy. However, PV power generation is affected by meteorological factors and has strong volatility, randomness and intermittency [2], which can affect the stability and safety of the entire power system after being connected to the grid. If the output of PV power can be predicted accurately and quickly, it can significantly improve the stability and reliability of the power grid. Therefore, it is particularly important to study how to improve the accuracy and timeliness of PV power prediction.

PV power prediction methods can be categorized into three types: physical methods, statistical methods and AI methods. Physical methods do not require a large amount of historical data but lack robustness. Statistical methods perform poorly in handling non-linear data [2]. While AI methods show better performance and stronger robustness in analyzing complex non-linear data [3].

AI methods mainly refer to utilizing deep learning networks to learn from past data, capture the long-term dependencies existing in time series and make predictions. Common deep learning networks include convolutional neural networks (CNN) [4, 5], recurrent neural networks (RNN) and its variants, such as long short-term memory (LSTM) [6–8] and gated recurrent unit (GRU) [9, 10]. Unlike traditional RNN, LSTM introduces gating mechanisms, including input gates, forget gates and output gates, which can better control the flow of information and effectively handle long-term dependencies [8]. At the same time, LSTM solves the problems of vanishing gradients and exploding gradients that exist in traditional RNN [5, 8]. GRU introduces the concept of reset gates and update gates, which can better capture the relationships between different time steps in a sequence [9]. Compared to LSTM, GRU has a simpler structure and faster computation speed [6].

Decomposing time series before prediction can reduce its complexity, facilitate feature extraction and improve the accuracy of subsequent prediction. Therefore, in recent years, the prediction structure combining signal decomposition and deep learning networks has become a consensus. Maximal overlap discrete wavelet transform (MODWT) [11, 12] is a commonly used time-frequency domain decomposition method. Complete ensemble empirical mode decomposition with adaptive

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noise (CEEMDAN) [13–15], variational mode decomposition (VMD) [16–18] and singular value decomposition (SVD) [19] are frequency domain decomposition methods. In reference [20], the integrated frequency-domain decomposition method was utilized to handle photovoltaic power data, resulting in a significant improvement in prediction accuracy.

For time series with a high level of complexity, single-domain decomposition methods struggle to fully decompose them. In such cases, it is necessary to employ the integrated bidomain decomposition method. In [21], the DWT-EMD method for integrating bidomain decomposition effectively extracted information from both the time domain and frequency domain of the time series, further enhancing prediction accuracy. However, the integrated bidomain decomposition method often decomposes hundreds of mode components and learning and predicting these mode components with deep learning networks would consume a large amount of time and resources, thereby reducing the timeliness of predictions. Therefore, it is necessary to analyze the complexity of the mode components using sample entropy (SE) [13, 14] and make corresponding processing for mode components with different complexities to effectively shorten the prediction time.

There are two forms of prediction: point prediction and interval prediction. Point prediction is difficult to describe the uncertainty of PV power, while interval prediction is more reasonable [22] as it provides the expected output value of PV power with a certain level of confidence. The minimum value specifies the lowest power output, allowing the scheduling department to refer to the prediction results and integrate PV power into the real-time power system based on the system's acceptance capacity. This provides support for risk management, optimization control and supply-demand balance in the electricity market with PV power generation, ensuring power supply [23]. The commonly used interval prediction methods include bootstrap, Gaussian distribution estimation (GDE), kernel density estimation (KDE) [20, 23] and quantile regression (QR) [24, 25]. However, bootstrap and GDE are statistical interval generation methods based on a large amount of existing data and KDE tends to overfit or underfit when extreme values appear in the dataset. Therefore, QR is more universal, since it does not require distribution assumptions, has relatively loose conditions and can effectively estimate the uncertainty of time series forecasting.

Based on the above analysis, this paper proposes a short-term interval prediction method for PV power based on double sample entropy framework, integrated bidomain decomposition and deep learning network. Firstly, the time series data is decomposed into several S-components using the MODWT method in the time-frequency domain decomposition. The components with low complexity are selected using sample entropy and inputted into the deep learning network for learning and prediction. The components with high complexity undergo a secondary decomposition using the frequency domain decomposition method. Then, the components with low complexity are selected using sample entropy, integrated and inputted into the deep learning network for learning and prediction. The components with high complexity are separately learned

and predicted using the deep learning network. The prediction results of each component are reconstructed to obtain point predictions. Finally, quantile regression is used to obtain interval predictions based on the point predictions. Through case analysis, the proposed method in this paper demonstrates advantages in both point prediction and interval prediction stages. It outperforms other integrated bidomain decomposition deep learning methods that are not in the double sample entropy framework and single decomposition deep learning methods.

2 | THEORETICAL FOUNDATIONS

2.1 | Sample entropy

SE measures the complexity of a time series by quantifying the probability of generating new patterns in the signal. The larger the probability of generating new patterns, the higher the value of SE, indicating a greater complexity in the sequence. Conversely, a lower value of SE indicates a higher self-similarity in the sequence [26].

SE has two advantages: the calculation of sample entropy does not depend on the length of the data; SE exhibits better consistency, meaning that the impact of changing the parameters m and r on SE is the same. Principle reference [13].

2.2 | Maximal overlap discrete wavelet transform

MODWT is a method of discrete wavelet transform and it is a discrete implementation of the continuous wavelet transform. It is commonly used in time series analysis and signal processing for multi-scale decomposition and frequency domain analysis. The main characteristic of MODWT is maximal overlap, which provides better time and frequency resolution and enables capturing rapid changes in the signal. Compared to traditional discrete wavelet transform (DWT), MODWT typically has fewer redundant coefficients, which aids in extracting temporal signal features. It also possesses shift-invariance, reducing the influence of the starting point of the time series on subsequent predictions and minimizing latency [23]. Principle reference [11].

2.3 | Modal decomposition method

CEEMDAN, VMD and SVD are commonly used modal decomposition methods. References [19, 28], for the principles of VMD and SVD.

CEEMDAN is an improved empirical mode decomposition (EMD) method, which is a frequency domain decomposition method used for component separation and time-frequency analysis of non-linear and non-stationary signals. It addresses the issues of mode mixing and convergence in traditional EMD by adaptively introducing noise into the EMD process. The strength and distribution of the adaptive noise are adjusted based on the energy of the residual signal to gradually

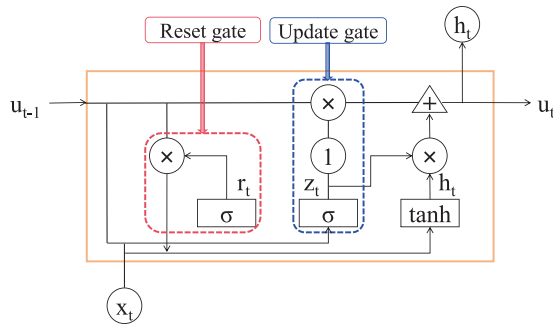


FIGURE 1 Gated recurrent unit neural cell structure diagram.

eliminate the residual noise. These noise perturbations are flat and random in the spectrum, which helps separate the local features in the original signal and reduce the interference between modes, resulting in better decomposition accuracy and faster decomposition speed [27]. Principle reference [13].

2.4 | Gated recurrent unit

GRU is another variant of the RNN, commonly used for modelling tasks involving sequential and time series data. Its neural cell structure is depicted in Figure 1. Compared to LSTM, GRU has a simpler structure, which leads to lower computational complexity and faster training speed [28]. The core idea of GRU is to use two gates to control the flow of information: the Reset Gate and the Update Gate. These gates help GRU decide how to use past information and where to retain the current information. GRU processes the entire sequence by repeating the steps mentioned in [10] and generates a new hidden state at each time step. Finally, the final hidden state can be used for task output or passed on to other network layers for further processing [9].

2.5 | Convolutional neural network

By introducing an attention mechanism, CNN-Attention significantly enhances the capabilities of CNN. It dynamically allocates attention weights to different spatial locations or channels, allowing the network to adaptively focus on the most salient regions. This enables prioritization and emphasis on relevant features while disregarding less informative or irrelevant parts [28]. This attention mechanism effectively enhances the discriminative power and accuracy of the network, improves the interpretability of predictions and enhances its ability to capture fine-grained details and subtle patterns in the input data. As a result, it achieves superior performance compared to traditional CNN models. Figure 2 shows the layer structure of CNN-Attention used here.

3 | METHOD STRUCTURE

The flowchart of the proposed prediction method in this paper is shown in Figure 3. It can be divided into two parts: point

prediction and interval prediction. The following sections will provide a detailed explanation of these two parts.

3.1 | Point prediction

Point prediction is the first part of the method, in which the integrated bidomain decomposition is used under the framework of double sample entropy and then deep learning networks are used to predict the PV power time series of the next day. The specific process is as follows:

- (i) The PV power time series is decomposed into several S-components using the time-frequency domain decomposition method, MODWT.
- (ii) The complexity of each S-component is calculated using sample entropy and a deep learning network is used to learn from the S-components with low complexity. The predicted results are denoted as S' .
- (iii) The S-components with high complexity are further decomposed into individual IMF-components using a frequency domain decomposition method.
- (iv) The complexity of each IMF-component is calculated using sample entropy and a deep learning network is used to learn from the IMF-components with high complexity.
- (v) The IMF-components with low complexity are integrated into an IMF-trend-component, which is then learned by a deep learning network. The predicted results from IMF and IMF-trend are denoted as IMF' .
- (vi) All the components are reconstructed to obtain other and all the components are reconstructed and inverse transformed to obtain the point prediction results.

After undergoing double sample entropy screening and integrated bidomain decomposition, the time series exhibits components with lower complexity that tend to be regular and the number of components is reduced. By using deep learning networks to learn and predict these components, the accumulation of prediction errors reduced significantly, resulting in enhanced prediction accuracy. Additionally, the learning and prediction time can be greatly shortened, improving the timeliness of the predictions.

3.2 | Interval prediction

The interval prediction is the second part of the method, mainly using quantile regression based on point prediction results to form a confidence interval. The ideal prediction intervals at different confidence levels can be obtained by selecting appropriate quantiles. This paper analyzes the relative error of the prediction results and adjusts the number of lags appropriately to better capture the short-term dependence of the data. The lag represents the number of time steps by which the observed variable or error term is delayed from the current time point and introduced into the model as an independent variable, making the model more robust.

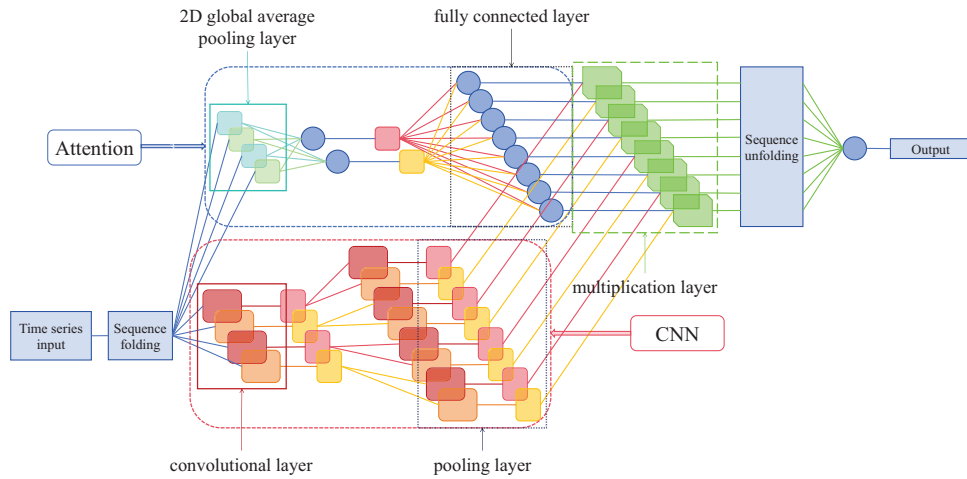


FIGURE 2 Convolution neural network-attention layer structure diagram.

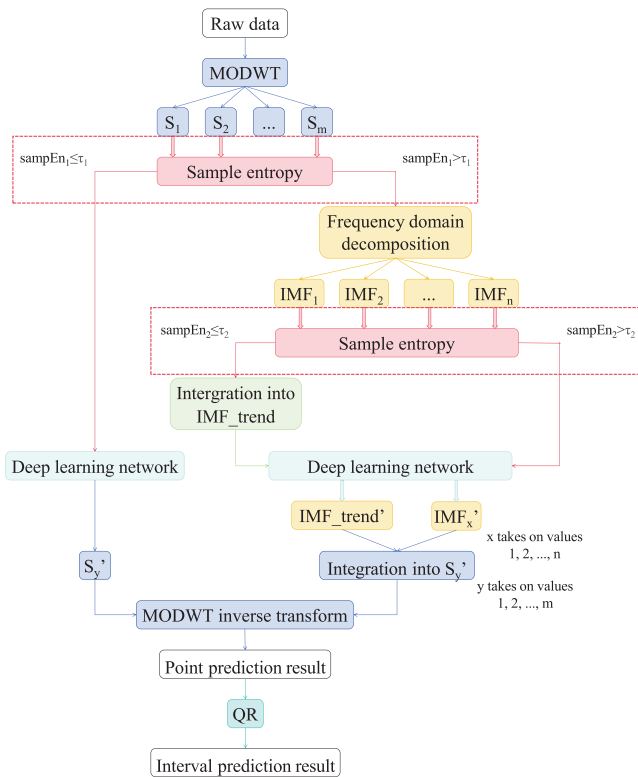


FIGURE 3 Flowchart of the prediction method.

3.3 | Analysis of the necessity of method structure

The method structure is necessary for addressing the challenges brought by the PV power time series data, such as the complexity and dynamics of the data. By cleverly combining methods such as bidomain decomposition, sample entropy analysis, deep learning and interval prediction, the accuracy, robustness and timeliness of the predictions are improved.

The use of bidomain decomposition allows for a more accurate representation of the underlying components in the PV

power time series. By decomposing the time series into S-components and IMF-components, the method captures both high-frequency and low-frequency variations, providing a more comprehensive understanding of the data.

Sample entropy analysis calculates the complexity of each component. The double sample entropy screening process filters out components with high complexity, which are further decomposed into IMF-components. This step ensures that the method focuses on learning and predicting the most relevant and predictable components, leading to improved accuracy while significantly reducing learning time.

The use of deep learning networks enables the method to learn from the decomposed components and make predictions based on the learned patterns. Deep learning networks are known for their ability to capture complex relationships in data, making them suitable for predicting the PV power time series.

Interval prediction is achieved through the use of quantile regression based on point prediction results. This provides a range of possible values for the predicted PV power, taking into account the uncertainty associated with the predictions. The selection of appropriate quantiles helps in obtaining prediction intervals at different confidence levels, providing a comprehensive assessment of the uncertainty in the forecasts.

4 | EXPERIMENTS

All experiments in this paper were conducted using Matlab.

4.1 | Data source and evaluation metrics

The data used in this case study are historical data from a PV power station in California, USA, provided by the Smart Grid Problems Competitions (website address: <http://www.gecad.isep.ipp.pt/ERM-competitions/home/>). Here, data from seven consecutive days with similar patterns and clear weather conditions (average solar radiation of 2400 W/m²) were selected as the dataset, with a sampling interval of five minutes, resulting in

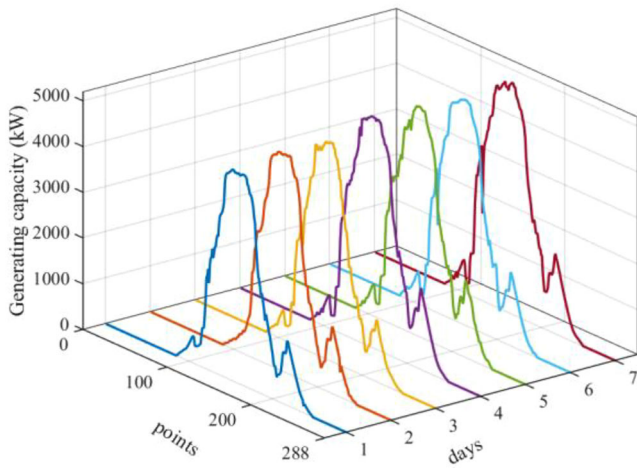


FIGURE 4 Resemblance days.

a total of 2016 data points, as shown in Figure 4. After undergoing integrated bidomain decomposition, the extracted features become more prominent. Therefore, theoretically, this method has lower data requirements.

In order to reasonably evaluate the rationality and accuracy of the proposed point predictions, three evaluation metrics are introduced: root mean square error (RMSE), mean absolute error (MAE) and mean absolute percentage error (MAPE).

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i)^2} \quad (1)$$

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |\hat{y}_i - y_i| \quad (2)$$

$$\text{MAPE} = \frac{1}{n} \sum_{i=1}^n \left| \frac{\hat{y}_i - y_i}{y_i} \right| \quad (3)$$

where $\hat{y}_i$, $\bar{y}_i$ and y_i are predicted value, mean value and actual value.

In order to examine the reliability and uncertainty level of the prediction intervals, this paper selects two comprehensive metrics: coverage width-based criterion (CWC) and continuous ranked probability score (CRPS), to evaluate the performance of interval predictions.

Coverage width-based criterion:

$$\text{CWC} = \text{PINAW}(1 + \gamma e^{-\eta(\text{PICP}-\mu)}) \quad (4)$$

where PINAW, η , μ and PICP are degree of narrowness within the interval, punishment factor, confidence level and interval coverage rate; γ is a Boolean value, equals 1 when the interval coverage rate is greater than or equal to the confidence level, otherwise equals 0.

continuous ranked probability score:

$$\text{CRPS}(F(x), P) = \int_{-\infty}^{+\infty} [F(x) - I_{\{x \geq P\}}]^2 dx \quad (5)$$

TABLE 1 Hyperparameter settings and values.

Hyperparameter settings	Values
White noise standard deviation multiplier	0.02
Decomposition iteration count	200
Number of hidden units (LSTM and GRU)	200
Number of convolutional kernels (CNN)	32 and 64
Solver	Adam
Loss function	MAE
Initial learning rate	0.005
Multiplicative factor	0.2
Training iteration count	200
Multiplicative epoch count	125

where $F(x)$, P and $I_{\{x \geq P\}}$ are cumulative distribution function, actual value and benchmark function.

4.2 | Hyperparameter settings and values in the method

The hyperparameter settings have a significant impact on the predictive performance of the method. The various algorithm hyperparameters mentioned in the article are set based on empirical values, as detailed in Table 1.

4.3 | Data preprocessing and analysis of training process

Taking the MODWT-CEEMDAN-GRU (SE) experiment as an example, where τ_1 is taken as 0.7Γ (Γ is a reference value). The data from the first to the sixth day is input and decomposed into 11 S-components through MODWT, as shown in Figure 5. Through sample entropy analysis, it is found that S_1 to S_4 exhibit higher complexity compared to other S-components. Therefore, these components are sequentially subjected to CEEMDAN decomposition, resulting in a varying number of IMF-components. The S_5 to S_{11} components are sequentially input into GRU for learning and prediction, obtaining the respective S' . Figure 6 displays the 13 IMF-components obtained after CEEMDAN decomposition of S_2 . Through sample entropy analysis, it is observed that IMF₁ to IMF₃ and IMF₅ to IMF₈ exhibit higher complexity compared to other IMF-components, indicating that these components need to be separately learned by GRU to obtaining the respective IMF'. By summing all the other IMF components, the IMF-trend component is obtained and input into GRU for learning and prediction, resulting in the respective IMF-trend'. The level of complexity is reflected in the decomposition graph as the strength or weakness of regular patterns.

Figure 7 shows the training process of GRU on the IMF₄. After more than 100 iterations, the loss function MAE

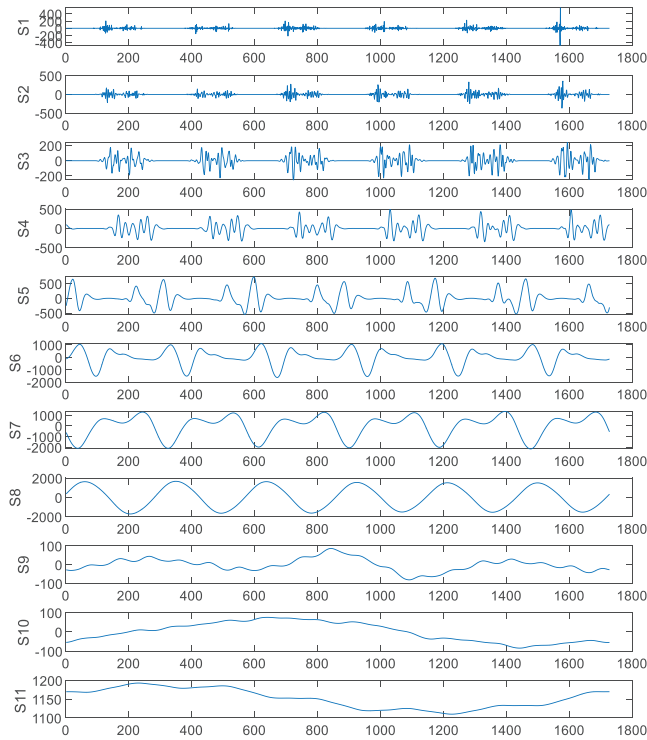


FIGURE 5 S-components.

stabilizes. The training loss approaches zero, indicating good training performance.

4.4 | Analysis of deep learning network adaptability

This section of the experiment selects the MODWT-CEEMDAN integrated bidomain decomposition under the framework of double sample entropy to test the adaptability of various deep learning networks. Here, τ_1 is taken as 0.7F. Table 2 shows the comparison of predictive performance of different deep learning networks. From Table 1, it can be observed that choosing GRU as the deep learning network has a slight advantage over choosing LSTM, CNN-Attention, GRU-LSTM and GRU-CNN-Attention, with an average reduction of 14.09%, 13.60% and 7.71% in point prediction error indicators RMSE, MAE and MAPE respectively.

(Note: The horizontal axis is in units of points and the vertical axis is in units of kW.)

It should be noted that:

- (i) Due to the limited number of samples selected in this study, the advantage of LSTM in capturing long-term dependencies is not evident, which does not mean that GRU is always superior to LSTM in any situation.
- (ii) Figures 8 and 9 show the correlation between different meteorological factors and PV power obtained through Pearson and Spearman correlation analysis. Among them, temperature and radiation are the meteorological factors

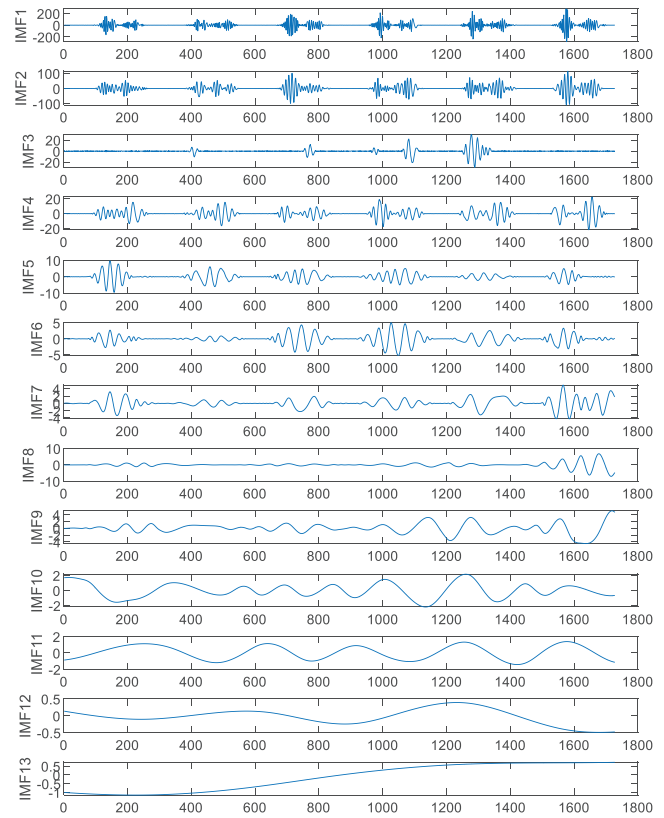


FIGURE 6 IMF-components.

with a correlation absolute value reaching 0.3. It is generally considered that correlation coefficients with an absolute value less than 0.3 are weakly correlated. Therefore, CNN-Attention only takes these two meteorological factors and the time series to form a multi-input single-output network structure.

- (iii) GRU-LSTM and GRU-CNN-Attention are horizontally combined deep learning networks, which means using GRU for training and prediction on S-components and IMF_trend-components, and using LSTM or CNN-Attention for training and prediction on IMF-components. Considering that IMF-components are still complex modal components after bidomain decomposition, this study attempts to use LSTM with more complex memory states and CNN-Attention with multiple inputs and added meteorological factor conditions to seek higher prediction accuracy. The former has poor results, while the latter, after adding meteorological factor conditions, has a smoother curve for point prediction, resulting in larger errors. Some indicators are better than GRU in interval prediction, but training CNN-attention takes longer time, and the timeliness of prediction is not ideal.
- (iv) Considering the superior learning and prediction performance of GRU on the dataset selected in this study, GRU is used for prediction in all subsequent experiments.

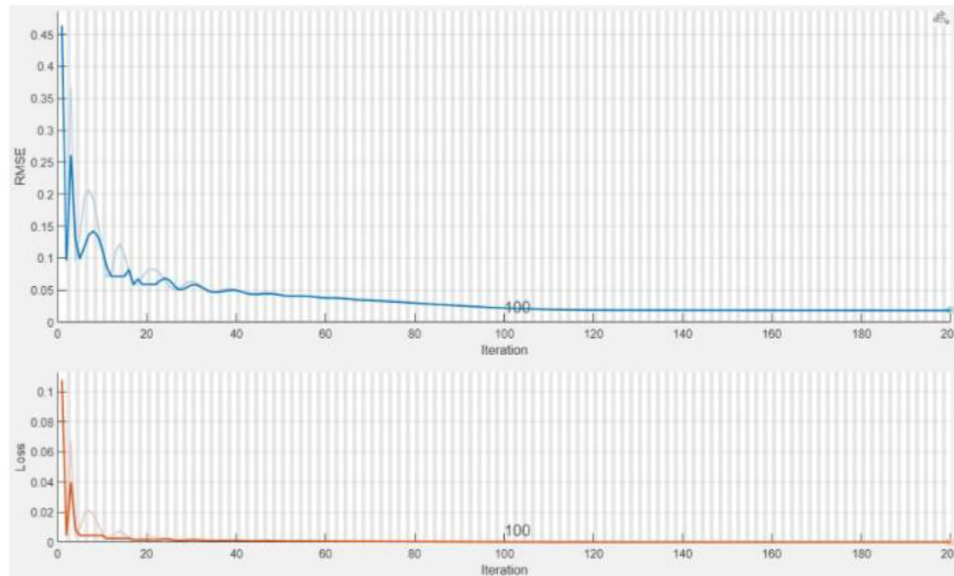


FIGURE 7 Training process of gated recurrent unit.

TABLE 2 Performance indicators of different deep learning networks.

Network Indicator	GRU	LSTM	CNN-attention	GRU-LSTM	GRU-CNN- attention
RMSE	1.1930	1.2633	1.7609	1.2458	1.2847
MAE	0.9911	1.0305	1.5154	1.0095	1.0333
MAPE	3.71%	3.82%	4.63%	3.77%	3.86%
90% confidence					
CWC	48.65%	62.68%	82.72%	47.10%	51.64%
CRPS	1.3557	1.3657	1.7320	1.3545	1.3448
70% confidence					
CWC	30.92%	32.57%	25.34%	29.94%	32.82%
CRPS	0.9682	0.9919	1.4043	0.9701	0.9387
50% confidence					
CWC	28.13%	37.97%	29.06%	31.55%	21.36%
CRPS	0.7094	0.7642	1.1159	0.7198	0.7569
Training and prediction time (s)	388	431	1251	429	902

4.5 | Integrated bidomain decomposition ratio analysis

After the time series undergoes integrated bidomain decomposition methods, the feature extraction becomes more pronounced, which is conducive to improving the prediction accuracy. However, at the same time, more modal components will add more errors to the prediction results. Therefore, this article experimented with three integrated bidomain decomposition methods and found the most suitable integrated bidomain decomposition ratios, as shown in Table 3. Considering the timeliness of prediction, the best secondary decomposition

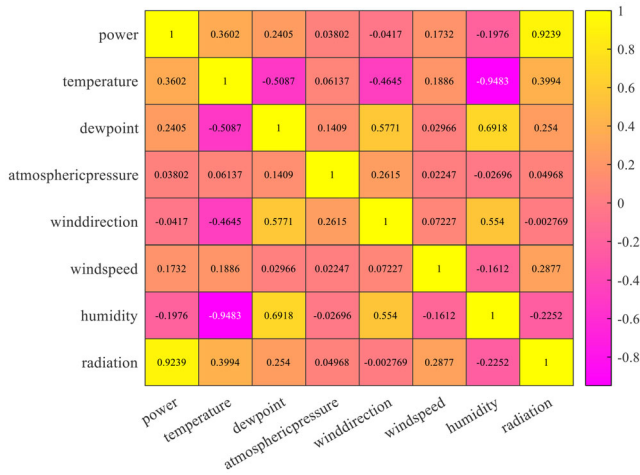
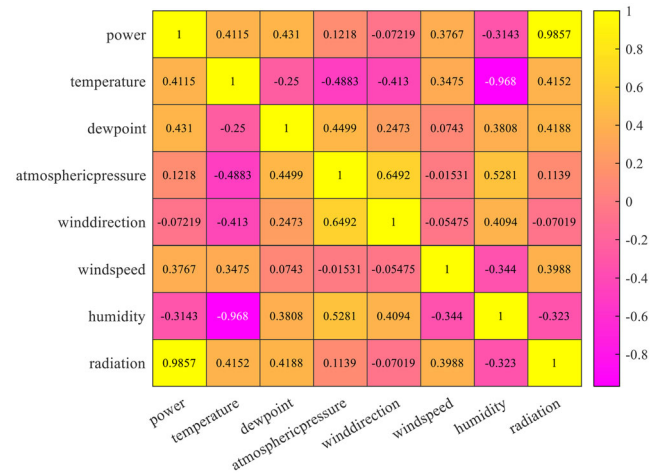
ratios of MODWT-CEEMDAN-GRU(SE), MODWT-AVMD-GRU(SE) and MODWT-SVD-GRU(SE) selected for the dataset in this study are 0.364, 0.182 and 0.091 respectively.

4.6 | Analysis of the advantages and disadvantages of each combination decomposition method

Under the framework of double sample entropy, each integrated bidomain decomposition method exhibits different strengths. The red sections in Table 3 represent the predictive

TABLE 3 Performance indicators of different decomposition ratios.

Integrated bidomain decomposition	MODWT-CEEMDAN (SE)			MODWT-AVMD (SE)				MODWT-SVD (SE)		
	Γ	0.8Γ and 0.7Γ	0.5Γ	Γ	0.8Γ	0.7Γ	0.5Γ	Γ and 0.8Γ	0.7Γ	0.5Γ
Decomposition ratio	0.091	0.364	0.455	0.091	0.182	0.364	0.455	0.091	0.273	0.364
RMSE	1.2259	1.193	1.1984	1.2082	1.1821	1.1768	1.1769	1.218	1.2189	1.2174
MAE	1.0765	1.0619	1.0647	1.0722	1.0558	1.0537	1.0549	1.0746	1.0752	1.0738
MAPE	3.77%	3.71%	3.72%	3.74%	3.68%	3.67%	3.68%	3.76%	3.76%	3.76%
90% confidence										
CWC	49.34%	48.65%	48.14%	41.22%	48.59%	48.74%	48.98%	49.19%	49.32%	49.75%
CRPS	1.3775	1.3557	1.3639	1.3628	1.3446	1.3413	1.3299	1.3589	1.3619	1.3572
70% confidence										
CWC	31.36%	30.92%	30.59%	26.23%	30.84%	30.94%	31.07%	31.26%	31.35%	31.62%
CRPS	0.9902	0.9682	0.9756	0.9819	0.9508	0.9472	0.9433	0.9777	0.9811	0.9787
50% confidence										
CWC	28.38%	28.13%	31.60%	34.70%	40.72%	40.77%	40.91%	20.35%	20.40%	20.58%
CRPS	0.7146	0.7094	0.7083	0.722	0.6902	0.6903	0.6912	0.7217	0.7214	0.7199
Training and prediction time /s	199	388	408	223	335	545	704	182	393	618

**FIGURE 8** Pearson correlation heat map.**FIGURE 9** Spearman correlation heat map.

performance of each integrated bidomain decomposition GRU method at their respective optimal secondary decomposition ratios.

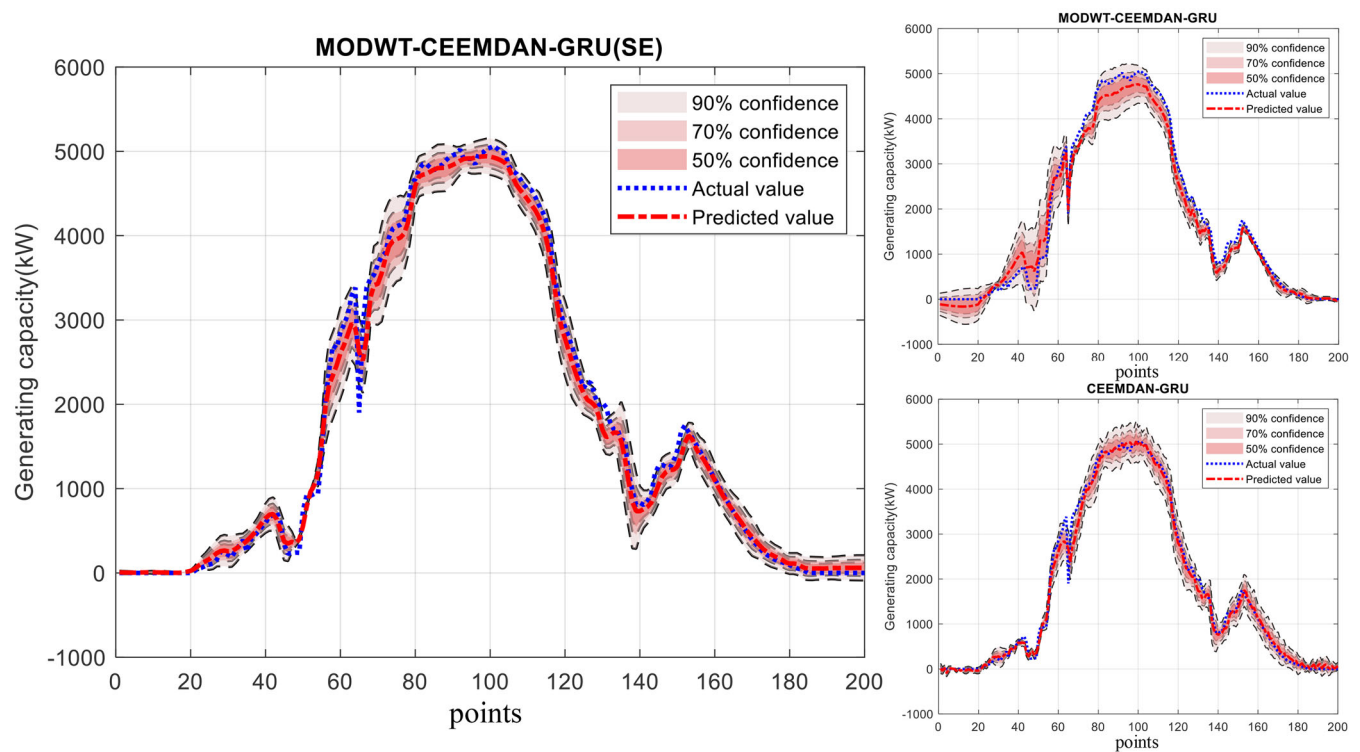
It can be inferred that MODWT-AVMD-GRU(SE) performs best in point forecasts and 90% and 70% confidence intervals, but its actual confidence level does not reach the required level for 50% confidence interval predictions. MODWT-SVD-GRU(SE) performs best in 50% confidence interval predictions and has the shortest prediction time. Although MODWT-CEEMDAN-GRU(SE) has the longest prediction time, all performance measures are comparatively stable, which makes it the most stable method.

4.7 | Prediction results analysis

In this section, the prediction methods of the integrated bidomain decomposition deep learning and the single decomposition deep learning were compared with the prediction method of integrated bidomain decomposition deep learning under the framework of double sample entropy proposed in this article (due to the large number of modal components decomposed by SVD-related methods, poor timeliness of prediction, hence not recorded). Table 4 shows the evaluation indicators of the prediction results of each method. Taking the three methods that use CEEMDAN as an example, Figure 10 shows a

TABLE 4 Performance indicators of different methods.

Indicator \ Method	GRU	CEEMDAN-GRU	AVMD-GRU	MODWT-CEEMDAN-GRU	MODWT-AVMD-GRU	MODWT-CEEMDAN-GRU (SE)	MODWT-AVMD-GRU (SE)	MODWT-SVD-GRU (SE)
RMSE	1.8532	<i>1.8408</i>	1.3422	<i>2.8356</i>	2.5854	<i>1.1930</i>	1.1821	1.2180
MAE	1.2349	<i>1.3946</i>	1.1168	<i>3.0521</i>	2.6711	<i>0.9911</i>	0.9854	1.003
MAPE	4.99%	<i>4.57%</i>	4.99%	<i>7.47%</i>	6.06%	<i>3.71%</i>	3.68%	3.76%
90% confidence								
CWC	40.80%	<i>18.77%</i>	35.45%	<i>93.28%</i>	40.38%	<i>48.65%</i>	48.59%	49.19%
CRPS	1.6299	<i>1.4306</i>	1.1716	<i>1.8798</i>	1.7175	<i>1.3557</i>	1.3446	1.3589
70% confidence								
CWC	15.92%	<i>27.33%</i>	45.49%	<i>56.85%</i>	51.68%	<i>30.92%</i>	30.84%	31.26%
CRPS	1.4073	<i>1.1183</i>	1.1307	<i>1.6712</i>	1.5782	<i>0.9682</i>	0.9508	0.9777
50% confidence								
CWC	10.36%	<i>36.26%</i>	28.93%	<i>38.32%</i>	34.46%	<i>28.13%</i>	40.72%	20.35%
CRPS	1.1138	<i>1.0378</i>	1.1273	<i>1.7083</i>	1.5432	<i>0.7094</i>	0.6902	0.7217
Training and prediction time (s)	11	<i>134</i>	170	<i>1450</i>	1901	<i>388</i>	335	182

**FIGURE 10** Comparison chart of prediction results.

comparison of their prediction results and 200 points covering the entire daytime portion were selected in the figure. The blue sections in Table 4 represent the predictive performance of the three. Compared with the other two methods, the method proposed in this article significantly improved the accuracy of

point prediction, reducing the error indicators RMSE, MAE and MAPE by an average of 45.56%, 48.23% and 34.58%, respectively. The interval predictions are more reasonable, with evaluation indicators CWC and CRPS reducing an average of 20.46% and 31.42%, respectively.

5 | CONCLUSION

In order to utilize photovoltaic energy better and address the fluctuations, randomness, intermittency and timeliness of PV power prediction, a PV power interval prediction method based on double sample entropy framework and integrated bidomain decomposition deep learning is proposed. Through case analysis, the following conclusions are drawn:

- (i) By incorporating sample entropy, the number of modal components obtained through integrated bidomain decomposition for the time series is reduced. This significantly reduces the computational cost of prediction, improves the timeliness of predictions and fundamentally reduces the issue of error accumulation in predictions. As a result, it enhances the accuracy of predictions.
- (ii) The integrated bidomain decomposition method under the framework of double sample entropy, can still effectively extract information from both the time and frequency domains in the time series. This enhances the learning effectiveness of the deep learning network and improves its robustness.
- (iii) The three integrated bidomain decomposition deep learning methods proposed here, combining sample entropy, exhibit advantages in predicting results under different circumstances. Depending on the actual situation, it is possible to flexibly select the appropriate method.
- (iv) The actual case studies demonstrate that compared to other methods, the proposed method achieves an average reduction of 45.56%, 48.23% and 34.58% in point prediction errors for RMSE, MAE and MAPE, respectively. Furthermore, in terms of interval prediction errors, there is an average reduction of 20.46% for CWC and 31.42% for CRPS.

In the author's opinion, for optimizing the methods proposed in this article, the approach mentioned in [29, 30], which focuses on optimizing the structure of neural networks based on the time correlation of new energy output, may become the mainstream optimization method in the future. Compared to parameter optimization algorithms that require significant time and computational resources to iteratively search for appropriate parameters to improve the learning effect of neural networks, this method can fundamentally enhance prediction accuracy by optimizing the neural network structure from the perspective of time correlation in new energy output.

AUTHOR CONTRIBUTIONS

Houkang Fang: Formal analysis; methodology; software; supervision; visualization; writing—original draft. **Qinmu Wu:** Conceptualization; funding acquisition; project administration; writing—review and editing. **Yihui Zhang:** Data curation; investigation; resources; validation.

CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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